

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1>
3. Khan Academy, Compound Circuits:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power>
4. EEVBlog, Voltage Dividers:
https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
<https://www.youtube.com/watch?v=WBfAEeEzDlq&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=13>

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

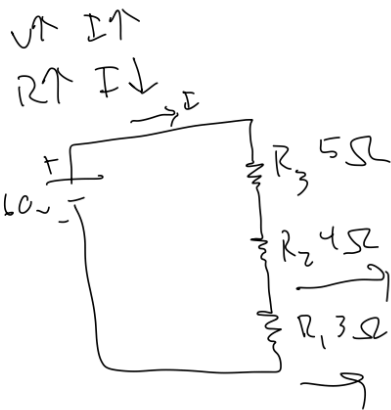
1. MIT OpenCourseWare Power Electronics Course:
 - a. [Lecture 5: Intro to DC/DC, Part 1](#)
2. [Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic](#)
3. [Power Electronics by Daniel W. Hart](#)

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Power electronics learning notes:

Ohms Law (organic chemistry tutor)

Ohms Law: $V = IR$
 $\uparrow \quad \uparrow \quad \leftarrow$
 volts amps ohms



Sum of voltage drops = battery voltage

$$V_1 = I R_1$$

$$15V = 5 \cdot 3$$

$$V_2 = I R_2$$

$$5 \cdot 4 = 20V$$

$$V_3 = I R_3$$

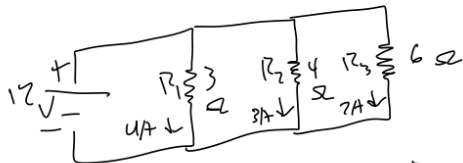
$$5 \cdot 5 = 25V$$

$$15 + 20 + 25 = 60V!$$

$$\frac{1}{3} + \frac{1}{4} = \frac{7}{12} + \frac{1}{6} = \frac{9}{12} = \frac{4}{3}$$

$$12 = \frac{4}{3} I$$

$$9 = I$$



$$V_1 = I_1 R_1$$

$$12 = 3 I_1$$

$$I_1 = 4A$$

$$V = I_2 R_2$$

$$12 = 4 I_2$$

$$I_2 = 3A$$

$$V = I_3 R_3$$

$$12 = 6 I_3$$

$$I_3 = 2A$$

$$4 + 3 + 2 = 9$$

$$R \uparrow I \downarrow$$

$$R \downarrow I \uparrow$$



$$R_T = R_1 + R_2 + R_3$$

$$R_T = 3 + 4 + 5 = 12$$

$$60V = 12 I$$

$$I = 5 \text{ amps}$$

$$12 = 4 I$$

$$I = 3 \text{ amps}$$

Khan Academy video:

(Volts)

Voltage (V) - Electric Potential

(Amps)

PE per unit charge

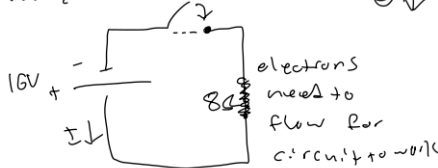
current (I) - how much

charge per unit time charge flowing past a point in circuit

Resistance (R) (ohms) in a second

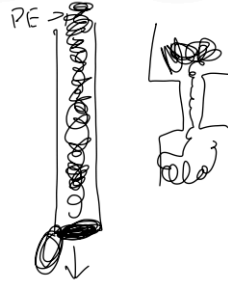
$V = IR$ how much charge flow impeded

$$I = \frac{V}{R}$$

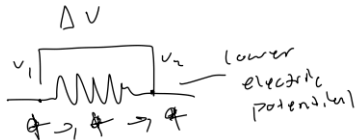


$$10 = 8I$$

$$I = 2 \text{ Amps}$$



Second Khan Academy Vid:



$$V = \frac{U}{q}$$

more PE than ending with decreasing electrons potential energy. speed is not increasing as current must be the same on both sides of the resistor. Energy is lost to heat, heating up the resistor as PE decreases

$$P = I^2 R \text{ also work!}$$

$$P = \frac{\Delta V^2}{R}$$

$$U_e = qV$$

$$qV_2 - qV_1 = \Delta U_e$$

$$P = \frac{E}{t} = \frac{\Delta U_e}{t} = q \frac{(V_2 - V_1)}{t}$$

$$= \frac{q \Delta V}{t}$$

$$P = \left(\frac{q}{t}\right) \Delta V$$

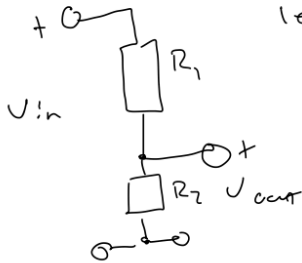
$$= I \Delta V \text{ watts}$$

$$\left(\frac{J}{s}\right)$$

works for anything that turns electric potential energy into some other kind of energy

Voltage Dividers; Used to get a known precise fraction of things to use in analog circuits

V_{out} must be less than V_{in}



$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{IN}$$

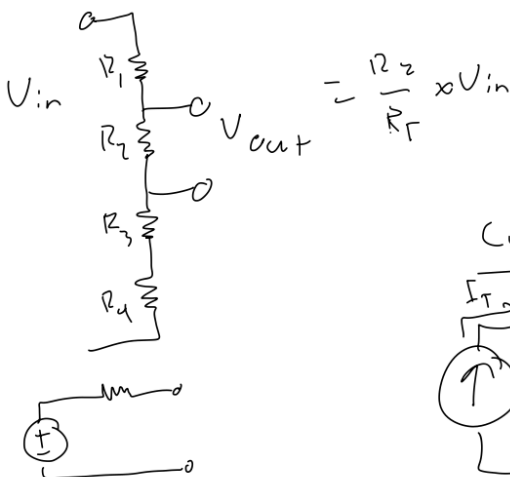
$$R_s = \sum R$$

$$\frac{1}{R_p} = \sum \frac{1}{R}$$

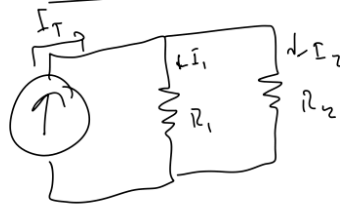
(for parallel)

Rules of thumb;

- For 10% lower cost use x10 value (look res in 1% x100 (1000k with 10k) parallel with 10k)
- 1% x1000 (10000k)
- $\frac{1}{2}$ 2 resistors in parallel // (same value)
- $\frac{1}{3}$ 3 ///



Current Dividers



$$I_1 = \frac{R_2}{R_1 + R_2} \times I_t$$

only for 2 resistors

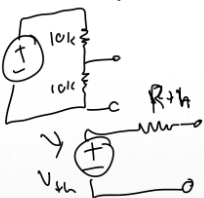
Voltage source

Current source

- Thevenin equivalent

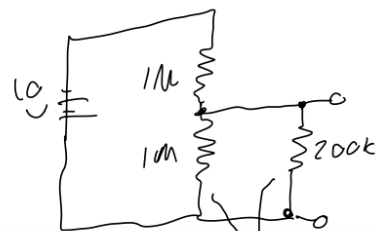
$$V_{th} = V_{open}$$

$$R_{th} = \frac{V_{open}}{I_{short}}$$



$$I_N = \frac{V_{th}}{R_{th}}$$

$$R_N = R_{th}$$



$$\frac{1}{\frac{1}{200k} + \frac{1}{1M}} = 166.6k$$

make it into one resistor

$$V_{th} = \frac{R_2}{R_1 + R_2} \times 10V = 5V$$

$$R_{th} = \frac{V_{th}}{I_{sh}} = \frac{5V}{1mA} = 5k$$

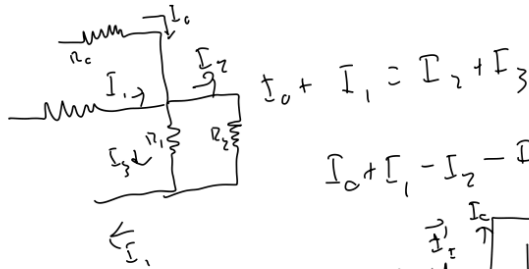
$$\frac{166.6k}{1M + 166.6k} \times 10V = 1.428V$$

Kirchoff's Laws:

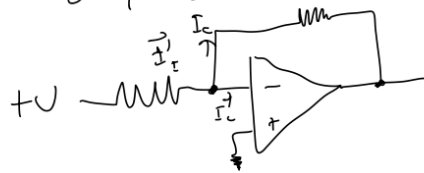
Current Law:

Sum of currents into a junction

= Sum of currents out of junction



$$I_0 + I_1 - I_2 - I_3 = 0$$



$$I_i = I_0 + I_1$$

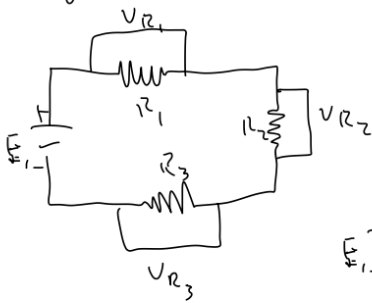
Voltage Law:

Sum of applied Voltages

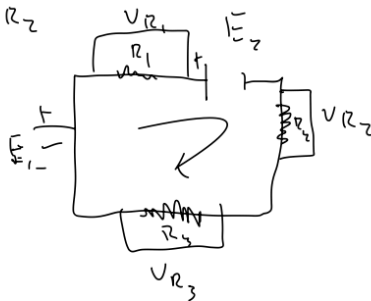
around a loop = Sum of voltage

drops around that loop

- $\rightarrow + = +$ for voltage
+ $\rightarrow - = -$ source



$$E_1 = V_{R1} + V_{R2} + V_{R3}$$



$$E_1 - E_2 = V_{R1} + V_{R2} + V_{R3}$$

$$E_1 = V_{R1} + V_{R2} + V_{R3} + E_2$$